

Innopolis University

Course: Big Data

User Behavior Analysis and Purchase Prediction in an Online Store

Final Project Report

Team: 18

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Date: May 11, 2026

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1 Introduction

Online retail platforms generate large volumes of user behavior data every day. Each customer interaction, such as viewing a product, adding an item to the cart, or completing a purchase, can provide valuable information about customer intent, product demand, and business performance. Analyzing such data at scale helps online stores improve marketing decisions, optimize inventory planning, reduce cart abandonment, and increase conversion rates.

The goal of this project is to build an end-to-end big data pipeline for analyzing ecommerce user behavior in an online store similar to Ozon. The project covers the full data processing cycle: data collection, relational storage, ingestion into HDFS, preparation of optimized Hive tables, exploratory data analysis, predictive modeling with Spark ML, and final presentation of results in Apache Superset.

The analytical objective of the project is twofold. First, we aim to understand how users interact with the online store across time, product categories, brands, sessions, and customer segments. Second, we formulate a predictive analytics task to estimate whether a user session will result in a purchase based on early-session behavior and historical user activity.

2 Business Objectives

The project focuses on business-oriented analysis of customer behavior in an online store. The main business objectives are:

1. Analyze when users are most active and when they are most likely to make purchases.
2. Identify product categories and brands that generate the highest purchase activity and customer loyalty.
3. Understand differences between passive browsers, engaged users, regular customers, and high-value customers.
4. Analyze session-level behavior and identify stages where users drop off before purchasing.
5. Detect high-intent time periods, such as weekend mornings, for more efficient marketing campaigns.
6. Build a predictive model that estimates whether a user session will end with a purchase.
7. Present the results in a dashboard that supports business decision-making.

3 Data Description

The dataset used in this project contains ecommerce event data from an online store. Each record represents a user action performed on the platform. The main event types include product views, cart additions, and purchases.

The dataset includes information about users, sessions, products, categories, brands, prices, event timestamps, and event types. This structure makes it suitable for both exploratory analysis and predictive modeling.

3.1 Dataset Source

The dataset was downloaded from Kaggle: <https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store>.

The download and extraction are automated in `scripts/data_collection.sh` using Kaggle API endpoint and `unzip`. The pipeline uses three monthly files (`2019-Oct.csv`, `2019-Nov.csv`, `2019-Dec.csv`), then loads them into PostgreSQL and then into HDFS/Hive.

3.2 Dataset Overview

Table 1: Dataset overview

Characteristic	Value
Dataset source	Kaggle: Ecommerce behavior data from multi-category store
Original file format	CSV (inside ZIP archive)
Number of records	177,493,585 events
Dataset size	7.71 GB in Hive Parquet (Snappy-compressed)
Date range	2019-10-01 to 2019-12-31
Number of features	9 raw event features
Number of unique users	7,856,637
Number of unique sessions	38,592,070
Number of unique products	239,718
Number of unique categories	1,215
Number of unique brands	4,846

3.3 Main Features

Table 2: Main dataset features

Feature	Type	Description
event_time	Datetime	Timestamp of the user event
event_type	Categorical	Type of event: view, cart, or purchase
product_id	Numerical / ID	Unique product identifier
category_id	Numerical / ID	Product category identifier
category_code	Categorical	Product category path or name
brand	Categorical	Product brand
price	Numerical	Product price
user_id	Numerical / ID	Unique user identifier
user_session	Categorical / ID	User session identifier

The staging table in PostgreSQL additionally contains a surrogate key (`event_id`) created as `BIGSERIAL PRIMARY KEY`. This key is used only for relational storage integrity and is not part of the original dataset schema.

4 Data Characteristics

Before performing analytical and predictive tasks, the dataset was explored to understand its structure, scale, and quality. This included checking the number of records, available features, event type distribution, missing values, and the number of unique entities such as users, sessions, products, categories, and brands.

4.1 Basic Characteristics

Table 3: Basic data characteristics

Metric	Value
Total number of events	177,493,585
Number of view events	167,321,573
Number of cart events	7,350,176
Number of purchase events	2,821,836
Number of unique users	7,856,637
Number of unique sessions	38,592,070
Number of unique products	239,718
Number of unique categories	1,215
Number of unique brands	4,846
Minimum event timestamp	2019-10-01 00:00:00
Maximum event timestamp	2019-12-31 23:59:59

4.2 Event Type Distribution

The event type distribution is important because it shows the structure of user behavior in the online store. Usually, product views dominate ecommerce event datasets, while cart additions and purchases represent smaller but more valuable user actions.

Table 4: Event type distribution

Event type	Number of events	Share, %
View	167,321,573	94.27
Cart	7,350,176	4.14
Purchase	2,821,836	1.59

4.3 Data Quality

Data quality checks were implemented at multiple stages. During PostgreSQL import (`sql/import_data`), rows with null critical fields (`event_time`, `event_type`, `product_id`, `category_id`, `user_id`, `user_session`) are filtered out in the `COPY` statement. In Spark (`scripts/model.py`), additional filtering removes records with invalid timestamps and keeps only supported event types (`view`, `cart`, `remove_from_cart`, `purchase`).

No explicit deduplication script is present in the repository; therefore, potential duplicates were preserved as behavioral signal. In the produced aggregates, `remove_from_cart`

counts are effectively zero, which should be treated as a dataset characteristic when interpreting funnel and abandonment metrics.

5 Architecture of the Data Pipeline

The project implements an end-to-end big data pipeline. The pipeline starts with raw e-commerce data and finishes with business insights and predictive analytics results presented in Apache Superset.

5.1 Pipeline Overview

The general architecture of the project is:

Raw Dataset → PostgreSQL/Citus → Sqoop → HDFS → Hive → EDA → Spark ML →
Superset Dashboard

5.2 Pipeline Stages

Table 5: Input and output of each pipeline stage

Stage	Input	Tool	Output
Data collection	Raw ecommerce data	Python / Bash	Prepared CSV files
Relational storage	CSV files	PostgreSQL + Citus	Distributed relational tables
HDFS ingestion	PostgreSQL tables	Sqoop	Data files in HDFS
Data warehouse	HDFS files	Hive	Optimized Hive tables
EDA	Hive tables	SQL on Tez	Aggregated analytical datasets
ML modeling	Hive tables	PySpark ML on YARN	Models, predictions, metrics
Presentation	EDA and ML outputs	Apache Superset	Final dashboard

5.3 Technology Stack

- PostgreSQL and Citus for distributed relational storage.
- Sqoop for importing relational data into HDFS.
- HDFS for distributed storage.
- Hive for data warehouse tables.
- Hive SQL on Tez for exploratory data analysis.
- PySpark and Spark ML for predictive analytics.

- Hadoop YARN for distributed Spark execution.
- Apache Superset for dashboard visualization.
- GitHub for code and project documentation.

6 Data Preparation

6.1 Relational Model

The raw ecommerce dataset was transformed into a relational structure suitable for PostgreSQL storage. The relational model was designed to represent the main business entities, such as events, users, sessions, products, categories, and brands.

The relational staging layer uses one transactional fact table: `events`. The table contains a surrogate primary key (`event_id`) and business columns from the raw dataset. A domain constraint is defined on `event_type` to allow only `view`, `cart`, `remove_from_cart`, and `purchase`.

The model intentionally keeps denormalized business attributes (`category_code`, `brand`, `price`) to simplify high-throughput ingestion and minimize join cost before Sqoop export to HDFS.

6.2 PostgreSQL and Citus

The prepared data was loaded into PostgreSQL. Citus was used to distribute relational tables across the database cluster. This allowed the project to store and process the dataset in a distributed environment before transferring it to HDFS.

The repository scripts directly create and fill PostgreSQL table `events` in database `team18_projectdb` (`scripts/build_projectdb.py`, `sql/create_tables.sql`, `sql/import_data.sql`). Connection endpoints are configured for the university cluster host (`hadoop-04.uni.innopolis.ru`).

Although Citus is listed in the technology stack, explicit `create_distributed_table(...)` statements are not included in the current codebase; operationally, the implemented flow relies on PostgreSQL staging followed by Sqoop ingestion into HDFS.

6.3 Import to HDFS with Sqoop

After loading the data into PostgreSQL, Sqoop was used to import the database tables into HDFS. The import process was automated through project scripts to make the results reproducible.

The imported data was stored in HDFS using Avro format and Snappy compression.

Sqoop import is automated in `scripts/data_ingestion.sh` with:

- **source:** `jdbc:postgresql://hadoop-04.uni.innopolis.ru/team18_projectdb`,
- **command:** `sqoop import-all-tables -as-avrodatafile -compression-codec=snap -compress -warehouse-dir=project/warehouse -m 1,`
- **target HDFS warehouse path:** `project/warehouse`.

Generated Avro schema artifacts are copied to `output/events.avsc` and `output/events.java`.

6.4 Hive Tables

Hive tables were created on top of the data stored in HDFS. The tables were optimized for analytical queries using efficient storage format, compression, and, where applicable, partitioning and bucketing.

Hive database name: `team18_projectdb`.

Table 6: Hive tables

Table name	Format	Compression	Optimization
<code>events_raw</code>	Avro	Snappy	External table over Sqoop-imported Avro files
<code>events_partitioned</code>	Parquet	Snappy	Partitioned by <code>year/month/day</code>
<code>session_features</code>	Parquet	Snappy	Partitioned by <code>session_date</code> , bucketed by <code>user_id</code> (16)

7 Exploratory Data Analysis

Exploratory Data Analysis was performed using SQL on the Tez engine. The purpose of EDA was to extract valuable business insights before data preprocessing and machine learning. Aggregated datasets were created in Hive and then visualized in Apache Superset.

The EDA part focuses on six major insights:

1. Daily shopping rhythm.
2. Product category performance.
3. Customer segmentation.
4. Brand loyalty and repeat buyers.
5. Session funnel analysis.
6. Weekday versus weekend purchase behavior.

7.1 Insight 1: Daily Shopping Rhythm

7.1.1 Business Question

At what time of day are users most active, and when are they most likely to make purchases?

7.1.2 Chart Description

This dashboard analyzes user activity by hour of day. It compares total events, conversion rate, average purchase value, and normalized event activity for views, cart additions, and purchases.

The chart is attached as a JPG artifact from dashboard exports.

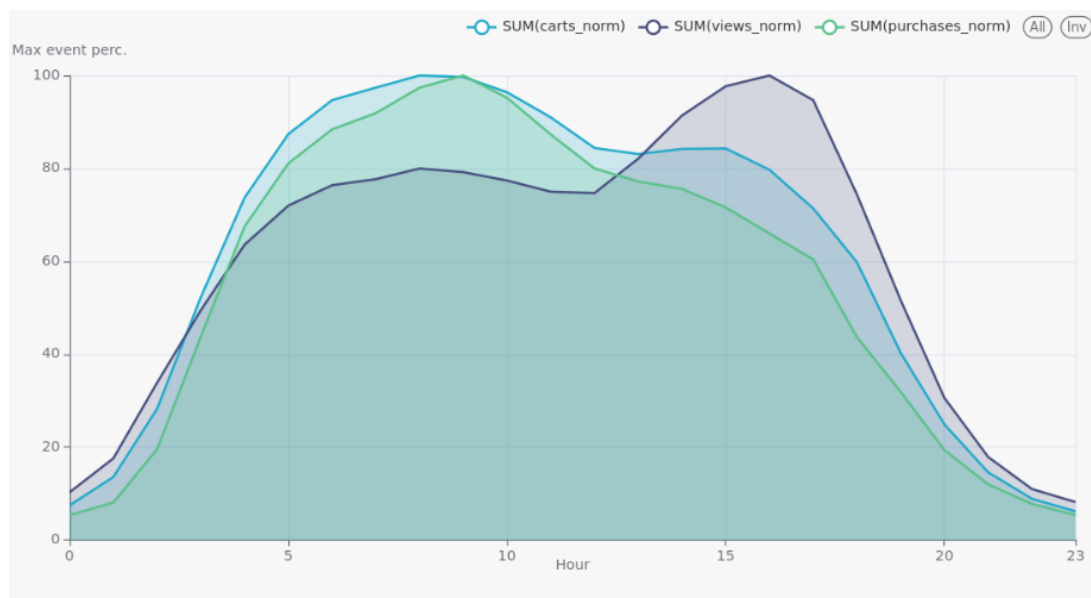


Figure 1: Daily shopping rhythm

7.1.3 Finding

User activity follows a clear daily rhythm. Browsing activity increases during the morning and reaches its highest level in the afternoon. However, purchase intent is stronger during earlier hours, while average purchase value tends to increase during evening hours.

7.1.4 Business Interpretation

High traffic does not always mean high purchase efficiency. Afternoon users may browse more actively, but morning users appear to be more conversion-oriented. Evening users may be fewer in number, but they can generate higher-value purchases.

7.1.5 Recommendation

Marketing campaigns aimed at conversion should be scheduled during morning and early afternoon hours. Evening hours can be used for premium product offers, bundles, and higher-value

promotions.

7.2 Insight 2: Product Category Performance

7.2.1 Business Question

Which product categories generate the most purchases, and which categories show strong conversion potential?

7.2.2 Chart Description

This dashboard compares product categories by number of purchases, cart-to-purchase rate, view-to-purchase rate, number of unique products, and number of unique users.

The chart is attached as a JPG artifact from dashboard exports.

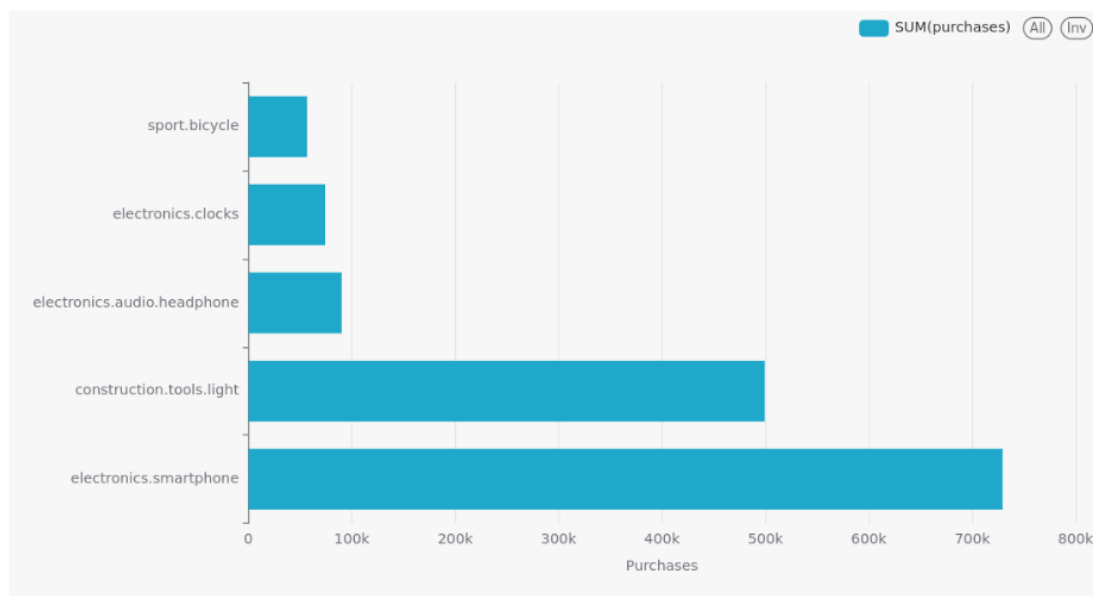


Figure 2: Product category performance

7.2.3 Finding

Purchases are highly concentrated in a small number of product categories. Some categories generate very large purchase volumes, while others have smaller traffic but strong cart-to-purchase efficiency.

7.2.4 Business Interpretation

The online store should distinguish between high-volume categories and high-efficiency niche categories. High-volume categories are important for revenue stability and inventory planning, while high-conversion niche categories may represent growth opportunities.

7.2.5 Recommendation

Inventory depth and marketing visibility should be protected for top-performing categories. At the same time, targeted campaigns should be tested for smaller categories with strong cart-to-purchase rates but limited traffic.

7.3 Insight 3: Customer Segmentation

7.3.1 Business Question

How do different customer segments behave, and which segments generate the most business value?

7.3.2 Chart Description

This dashboard groups users into behavioral segments such as passive browsers, active browsers, engaged browsers, regular customers, and high-value customers. It compares average sessions, active days, views, carts, purchases, explored categories, and conversion rate.

The chart is attached as a JPG artifact from dashboard exports.

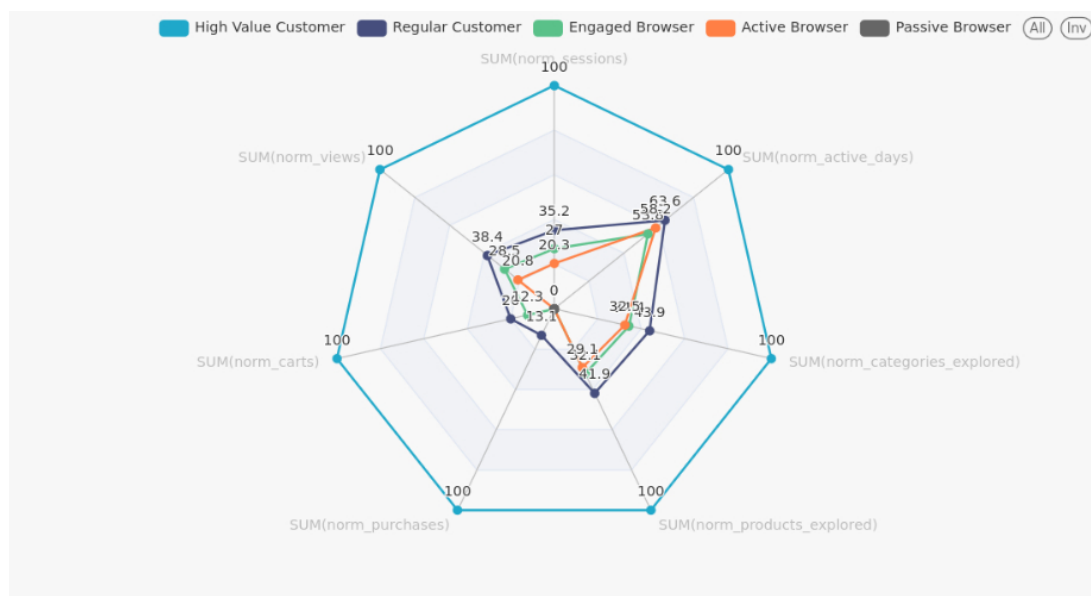


Figure 3: Customer segmentation

7.3.3 Finding

Conversion is concentrated among highly engaged and high-value users. Passive and browsing-oriented segments generate activity but show little or no purchase behavior. High-value customers demonstrate much stronger engagement and repeat purchase behavior.

7.3.4 Business Interpretation

The customer base is not homogeneous. Different user groups require different marketing strategies. Passive browsers need activation, engaged browsers need conversion support, and high-value customers need retention and loyalty programs.

7.3.5 Recommendation

The business should create separate lifecycle campaigns for each segment. Passive users should receive activation campaigns, engaged users should receive product and cart nudges, and high-value users should be targeted with VIP retention programs.

7.4 Insight 4: Brand Loyalty and Repeat Buyers

7.4.1 Business Question

Which brands have the strongest customer loyalty and repeat purchase behavior?

7.4.2 Chart Description

This dashboard analyzes brand-level performance using loyalty rate, repeat buyer count, unique users, conversion rate, and purchase activity.

The chart is attached as a JPG artifact from dashboard exports.

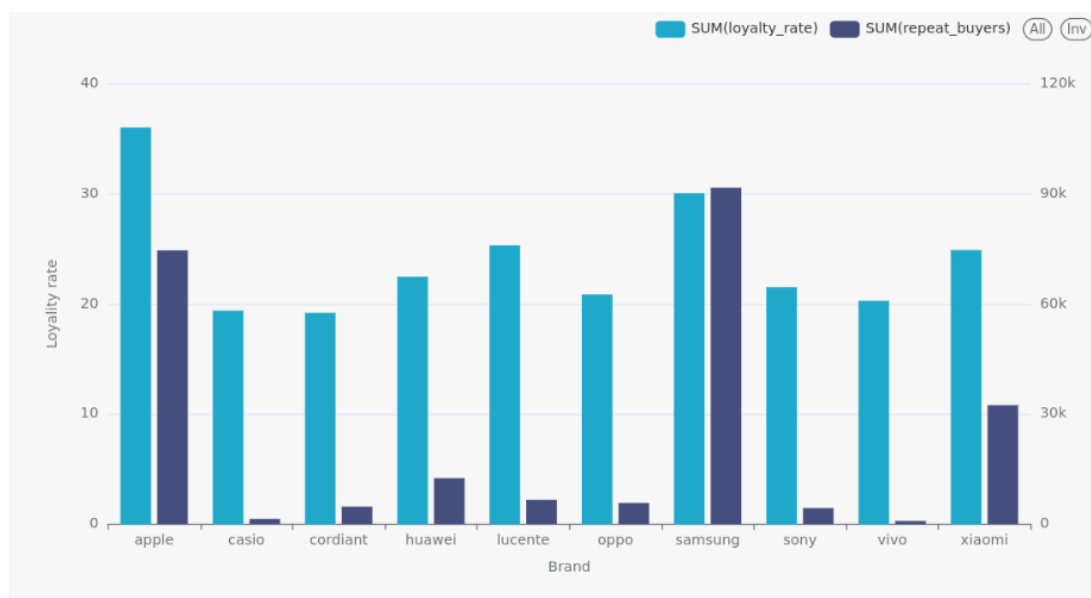


Figure 4: Brand loyalty and repeat buyers

7.4.3 Finding

Brand performance is uneven. Some brands show stronger loyalty rates, while others attract a larger number of repeat buyers. Premium and popular electronics brands appear to have stronger

repeat-purchase potential.

Top brands by purchases in the exported result are Samsung (638,864 purchases) and Apple (518,449 purchases). Apple also has the highest loyalty rate in top brands (36.02%), while Samsung combines the largest sales volume with 30.05% loyalty rate.

7.4.4 Business Interpretation

Strong brand loyalty can be used for partnership decisions, promotional planning, and customer retention. Brands with high repeat buyer counts are especially valuable because they indicate stable customer demand.

7.4.5 Recommendation

The online store should prioritize partnerships and retention campaigns with brands demonstrating high loyalty and repeat purchase metrics. Brand-specific offers, upgrade programs, accessories, and bundles can help increase customer lifetime value.

7.5 Insight 5: Session Funnel Analysis

7.5.1 Business Question

At which stages do user sessions drop off before purchase?

7.5.2 Chart Description

This dashboard presents a session funnel that groups sessions into bounce sessions, low engagement sessions, high engagement sessions, cart sessions without purchase, and purchase sessions.

The chart is attached as a JPG artifact from dashboard exports.

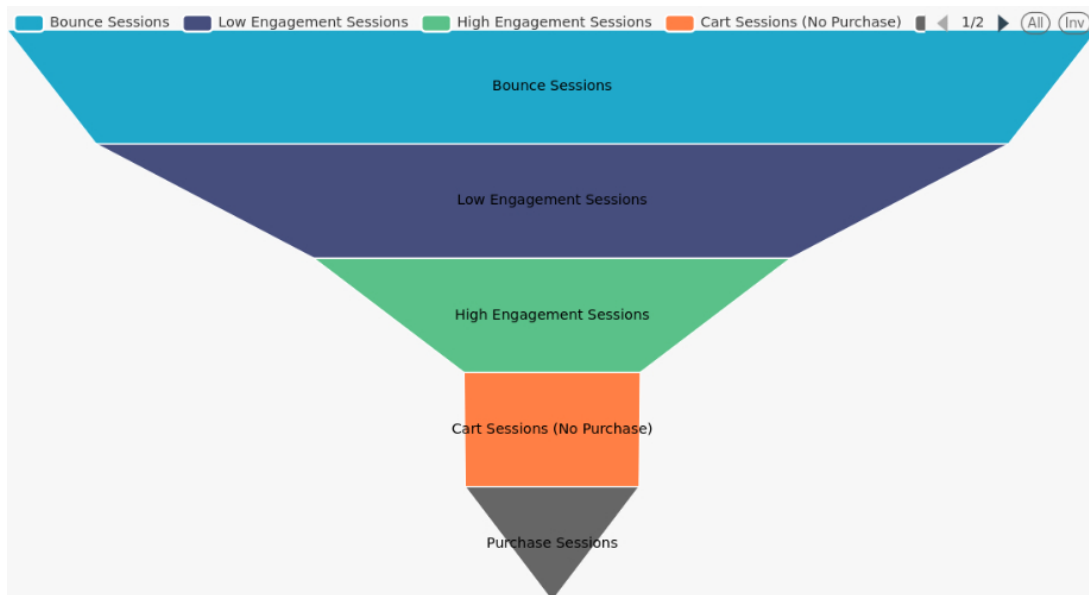


Figure 5: Session funnel analysis

7.5.3 Finding

A large share of sessions ends before meaningful engagement. Purchase sessions represent the smallest part of the funnel. Cart sessions without purchase form a critical near-conversion group.

7.5.4 Business Interpretation

Users who reach the cart but do not purchase are close to conversion. This group is especially important because small improvements in cart recovery can directly increase revenue.

7.5.5 Recommendation

The business should use two levels of optimization: reduce early-session drop-off through better landing pages and product discovery, and deploy real-time cart recovery triggers for users who add products to cart but do not complete the purchase.

7.6 Insight 6: Weekday vs Weekend Purchase Behavior

7.6.1 Business Question

How does purchase behavior differ between weekdays and weekends?

7.6.2 Chart Description

This dashboard compares purchase rate by hour of day for weekdays and weekends.

The chart is attached as a JPG artifact from dashboard exports.

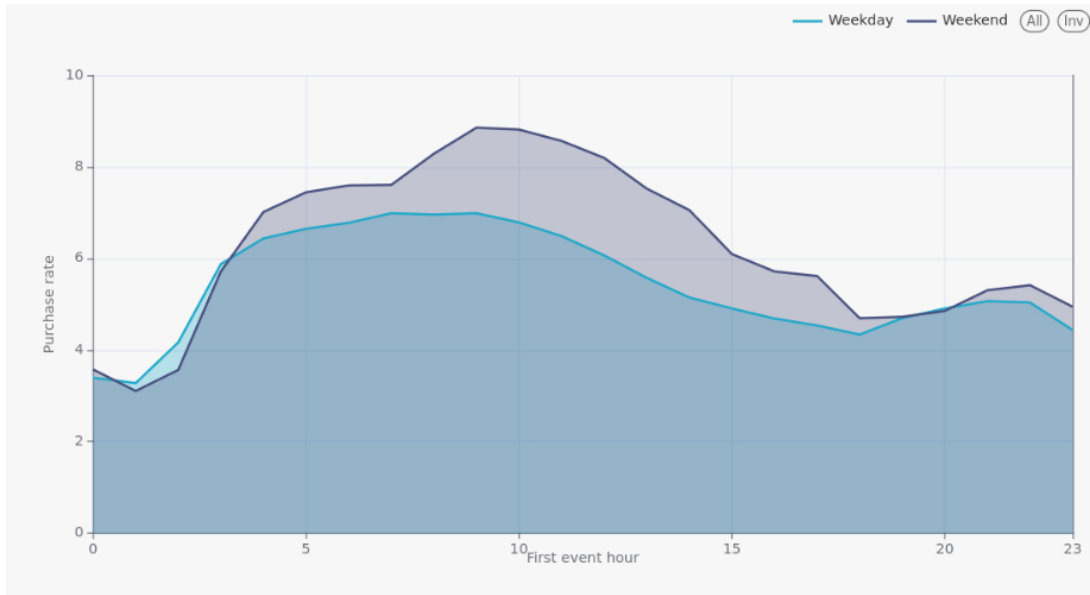


Figure 6: Weekday versus weekend purchase behavior

7.6.3 Finding

Purchase rate is higher during weekend mornings compared to weekday afternoon periods. The strongest purchase rate is observed during morning hours on weekends.

7.6.4 Business Interpretation

Weekend mornings represent a high-intent time window. Users during this period are more likely to complete purchases than users browsing during weaker weekday periods.

7.6.5 Recommendation

Advertising spend and flash sale events should be concentrated during weekend morning hours. Low-conversion weekday afternoon periods should receive reduced spend or be used mainly for retargeting campaigns.

8 Predictive Data Analysis

Predictive Data Analysis was performed using PySpark ML on the Hadoop YARN cluster. The main objective was to predict whether a user session would result in a purchase.

8.1 ML Task Definition

The predictive task was formulated as a binary classification problem.

- Label = 1: the session contains at least one purchase event.

- Label = 0: the session does not contain a purchase event.

The model uses early-session behavior and historical user features to estimate purchase probability.

8.2 Feature Engineering

The feature engineering process transformed raw ecommerce events into session-level features suitable for machine learning.

The following feature groups were used:

- Features extracted from the first 10 events of each session.
- Session-level activity features, such as number of views and cart additions.
- Time-based features, such as event hour and weekday/weekend indicators.
- Product, category, and brand interaction features.
- User-history features, such as previous purchases and previous session behavior.

The final model dataset (see `scripts/model.py`) includes:

- Prefix behavior counts: `views_count`, `carts_count`, `remove_from_cart_count`, `distinct_products`, `distinct_categories`, `distinct_brands`.
- Price and pace features: `avg_price`, `max_price`, `min_price`, `prefix_duration_seconds`, `avg_time_between_events_seconds`.
- Funnel ratio features: `cart_to_view_rate`, `remove_to_cart_rate`, `cart_in_first_3_events_f`.
- User history (30-day / inter-session): `user_purchase_count_30d`, `user_days_since_last_session`, `user_purchase_rate`.
- Temporal cyclical encodings: `first_event_hour_sin/cos`, `first_event_day_sin/cos`, `first_event_month_sin/cos`.
- Categorical prefix context: `last_prefix_event_type` (`StringIndexer` + `OneHotEncoder`).

8.3 Data Split

The dataset was split into training and test subsets. The training set was used for model fitting, grid search, and cross-validation. The test set was used only for final evaluation.

Table 7: Train-test split

Subset	Share	Purpose
Training set	70%	Training and cross-validation
Test set	30%	Final model evaluation

8.4 Model Selection

Following the Stage III Predictive Data Analytics checklist, two Spark ML model types were selected and compared:

1. Logistic Regression.
2. Random Forest Classifier.

Logistic Regression was used as a linear probabilistic baseline model. It is suitable for binary classification and provides interpretable purchase probabilities.

Random Forest was used as a non-linear ensemble model. It can capture complex relationships and interactions between session-level behavior and user-history features.

8.5 Hyperparameter Tuning

Both models were tuned using grid search and k-fold cross-validation. Hyperparameter optimization was performed only on the training data.

Cross-validation uses 3 folds (`numFolds=3`) with fixed seed (`SEED=18`) and `parallelism=2`.

8.5.1 Logistic Regression Hyperparameters

Table 8: Logistic Regression hyperparameter grid

Hyperparameter	Values
regParam	{0.01, 0.1}
elasticNetParam	{0.0, 0.5}
threshold	{0.35, 0.5}

8.5.2 Random Forest Hyperparameters

Table 9: Random Forest hyperparameter grid

Hyperparameter	Values
maxDepth	{5, 10}
maxBins	{32, 64}
subsamplingRate	{0.7, 1.0}

8.6 Evaluation Metrics

Since the task is binary classification, the main evaluation metrics were:

- Area Under ROC.
- Area Under PR.
- Accuracy.
- F1 score.

Area Under PR is especially important for this task because purchase sessions are expected to be less frequent than non-purchase sessions. Therefore, PR AUC provides a useful view of model quality on the positive purchase class.

8.7 Model Evaluation Results

Table 10: Model evaluation results

Model	ROC AUC	PR AUC	Accuracy	F1 Score
Logistic Regression	0.7662	0.2627	0.9249	0.9006
Random Forest	0.7845	0.3085	0.9283	0.9001

8.8 Best Model

Both models are trained and evaluated in `scripts/model.py`. Based on the updated metrics, Random Forest is the best model because it outperforms Logistic Regression in ROC AUC (0.7845 vs 0.7662), PR AUC (0.3085 vs 0.2627), and Accuracy (0.9283 vs 0.9249), while keeping a comparable F1 score.

The best model was selected based on the evaluation metrics on the test dataset. The selected model was saved to HDFS and its prediction results were exported for dashboard visualization.

8.9 Prediction Example

The project also includes prediction for a specific data sample.

Table 11: Input prefix events for the prediction example session

#	event_time	event_type	product_id	category_id	brand	price
1	2019-11-17T04:55:00	view	12200907	2116907525176557699	torrent	64.33
2	2019-11-17T04:55:16	view	12201355	2116907525176557699	torrent	64.69
3	2019-11-17T04:55:25	view	12201355	2116907525176557699	torrent	64.69
4	2019-11-17T04:55:41	view	12201246	2116907525176557699	racer	84.69
5	2019-11-17T04:56:09	view	12201687	2116907525176557699	axis	97.81
6	2019-11-17T04:56:38	view	12201687	2116907525176557699	axis	97.81
7	2019-11-17T04:57:06	view	12201688	2116907525176557699	axis	97.81
8	2019-11-17T04:57:21	view	12201688	2116907525176557699	axis	97.81
9	2019-11-17T04:57:31	view	12201687	2116907525176557699	axis	97.81
10	2019-11-17T04:57:44	cart	12201687	2116907525176557699	axis	97.81

The next events in the same session confirm the positive outcome (purchase after the first 10 events):

Table 12: Events after the 10-event prefix (ground truth)

#	event_time	event_type	product_id	category_id	brand	price
11	2019-11-17 04:58:00	purchase	12201687	2116907525176557699	axis	97.81
14	2019-11-17 04:59:13	purchase	12201688	2116907525176557699	axis	97.81

9 Data Presentation and Dashboard

The final results were presented in Apache Superset. The dashboard was designed to communicate both data characteristics and analytical findings to business stakeholders.

9.1 Dashboard Structure

The dashboard contains the following parts:

1. Data characteristics overview.
2. Daily shopping rhythm analysis.
3. Product category performance.
4. Customer segmentation.
5. Brand loyalty and repeat buyer analysis.

6. Session funnel analysis.
7. Weekday versus weekend purchase behavior.
8. Machine learning model performance.
9. Example prediction results.

9.2 Dashboard Purpose

The purpose of the dashboard is not only to display charts, but also to tell a business story based on user behavior. Each chart is accompanied by a short description, interpretation, and recommendation.

The dashboard helps answer the following business questions:

- When should marketing campaigns be scheduled?
- Which categories should receive more inventory and marketing support?
- Which customer segments should be activated or retained?
- Which brands demonstrate stronger loyalty?
- Where do users drop off before purchase?
- Can the business predict purchase sessions in advance?

Dashboard figures in this report are attached as exported JPG artifacts and stored in the repository

10 Findings

The main findings of the project are:

1. User traffic peaks do not always correspond to purchase peaks. Users browse actively during the day, but purchase efficiency is stronger during specific time windows.
2. Weekend mornings demonstrate stronger purchase behavior than weaker weekday periods.
3. Purchases are concentrated in a small number of product categories, while some smaller categories show high conversion potential.
4. High-value customers generate disproportionate business value through stronger engagement and repeated purchases.

5. Cart sessions without purchase represent a critical near-conversion group and should be targeted with recovery actions.
6. Brand loyalty differs significantly across brands, which creates opportunities for brand-specific partnerships and promotions.
7. Machine learning models can estimate whether a session will result in a purchase using early-session and user-history features.

11 Challenges and Difficulties

During the project, the team faced several technical and analytical challenges:

- Processing a large ecommerce dataset in a distributed environment.
- Designing a relational model suitable for PostgreSQL and Citus.
- Importing data from PostgreSQL into HDFS using Sqoop.
- Creating optimized Hive tables with appropriate storage format and compression.
- Writing SQL queries for business-oriented EDA insights.
- Preparing aggregated datasets for dashboard visualizations.
- Engineering session-level and user-history features for predictive modeling.
- Running Spark ML applications on the YARN cluster.
- Integrating EDA and ML outputs into Apache Superset.

Key implementation issues and resolutions:

- Dataset mirror instability: direct Rees46 links were unavailable, so ingestion was switched to Kaggle API download in `data_collection.sh`.
- Large-scale ETL orchestration: the pipeline was split into three reproducible stages (`stage1.sh`, `stage2.sh`, `stage3.sh`) with explicit logging.
- Storage/query performance: Hive tables were redesigned to Parquet + Snappy with partitioning (`events_partitioned`) and bucketing (`session_features`, `user_historical_features`).
- Feature leakage risk for ML: labels are built only from events after the first 10 prefix events to keep a realistic early-session prediction setup.

12 Business Recommendations

Based on the analysis, the following business recommendations can be made:

1. Schedule conversion-oriented marketing campaigns during high-intent time windows, especially morning hours and weekend mornings.
2. Use evening periods for premium offers, bundles, and higher-value product promotions.
3. Protect inventory and advertising visibility for top-performing product categories.
4. Test growth campaigns for smaller categories with strong cart-to-purchase efficiency.
5. Create activation campaigns for passive and active browsers.
6. Develop VIP retention programs for high-value customers.
7. Use real-time cart recovery triggers for users who add products to cart but do not complete purchase.
8. Prioritize brand partnerships with brands that demonstrate high loyalty and repeat buyer metrics.
9. Use purchase prediction scores to personalize offers and target users with high purchase probability.

13 Conclusion

This project implemented an end-to-end big data pipeline for ecommerce user behavior analysis and purchase prediction. The pipeline included data collection, relational storage in PostgreSQL with Citus, ingestion into HDFS with Sqoop, creation of optimized Hive tables, exploratory data analysis with SQL, predictive modeling with Spark ML on YARN, and dashboard presentation in Apache Superset.

The exploratory analysis showed that customer behavior is highly uneven across time, categories, brands, sessions, and user segments. Traffic peaks do not always match purchase peaks, and a small number of categories, brands, and high-value users generate a large share of business value. The session funnel analysis also showed that cart abandoners are a critical near-conversion group.

The predictive analytics part formulated session purchase prediction as a binary classification problem. Logistic Regression and Random Forest models were trained and evaluated using distributed Spark ML. The models used early-session activity and user-history features to estimate whether a session would result in a purchase.

Overall, the project demonstrates how big data technologies can be used to transform raw ecommerce events into actionable business insights and predictive tools for decision-making.

14 Reflections on Own Work

The strongest learning outcome was practical experience in connecting relational ingestion, HDFS/Hive warehousing, and Spark ML into one reproducible pipeline. The main area for improvement is artifact completeness in the repository snapshot (for example, exporting and versioning final Stage III metrics and dashboard screenshots together with the report to make the narrative fully self-contained).

During the project, the team gained practical experience with the full big data pipeline, including distributed storage, data warehouse preparation, analytical query design, Spark ML modeling, and dashboard-based storytelling.

15 Team Contributions

15.1 Role Responsibilities

- **Kirill Greshnov (ML Specialist):** prepared dataset for ML modeling, built distributed ML models, monitored model quality.
- **Denis Troegubov (Data Engineer):** managed databases, built the data pipeline, wrote automation scripts.
- **Vladimir Paskal (Data Scientist):** analyzed and understood the data, performed EDA, analyzed features, built dashboards, and prepared data storytelling.
- **Albert Shammassov (Tester and Tech Writer):** documented each project stage, organized the final report and repository documentation, tested project artifacts, and checked artifact quality.

Table 13: Team contribution table

Project Task	Task Description	Kirill	Denis	Vladimir	Albert	Deliverables	Avg. Hours
Data collection and ingestion	Automated data collection, created PostgreSQL database, loaded data, imported tables into HDFS using Sqoop, used Avro and Snappy.	10%	60%	15%	15%	Stage I scripts, PostgreSQL tables, HDFS files	18
Data warehouse preparation	Created optimized Hive tables with partitioning and bucketing for further analytics.	15%	50%	20%	15%	Hive tables, Stage II scripts	20

Project Task	Task Description	Kirill	Denis	Vladimir	Albert	Deliverables	Avg. Hours
Exploratory Data Analysis	Understood the data, performed EDA, analyzed features, and prepared business insights from dashboard charts.	10%	15%	60%	15%	EDA SQL scripts, aggregated datasets, charts	24
Predictive Data Analytics	Built Spark ML pipeline for session purchase prediction, engineered features, trained Logistic Regression and Random Forest models, evaluated models, and saved outputs.	65%	15%	10%	10%	ML scripts, saved models, predictions, evaluation results	26
Dashboard	Created Apache Superset dashboard with data characteristics, EDA insights, ML performance, and prediction results.	10%	15%	60%	15%	Superset dashboard	16
Report	Prepared final project report with pipeline description, EDA interpretation, ML results, findings, and recommendations.	10%	10%	20%	60%	Final report	22
Presentation	Prepared final presentation and dashboard demo for project defence.	20%	20%	40%	20%	Presentation PDF / slides	12

16 References

- Dataset source (Kaggle): <https://www.kaggle.com/datasets/mkechinov/ecommerce-behavior-data-from-multi-category-store>
- Apache Hadoop documentation: <https://hadoop.apache.org/>
- Apache Hive documentation: <https://hive.apache.org/>
- Apache Spark MLlib documentation: <https://spark.apache.org/mllib/>
- Apache Superset documentation: <https://superset.apache.org/>
- PostgreSQL documentation: <https://www.postgresql.org/docs/>
- Citus documentation: <https://docs.citusdata.com/>

- Apache Sqoop documentation: <https://sqoop.apache.org/>